2026 – 27 COMP4913 The Capstone Project

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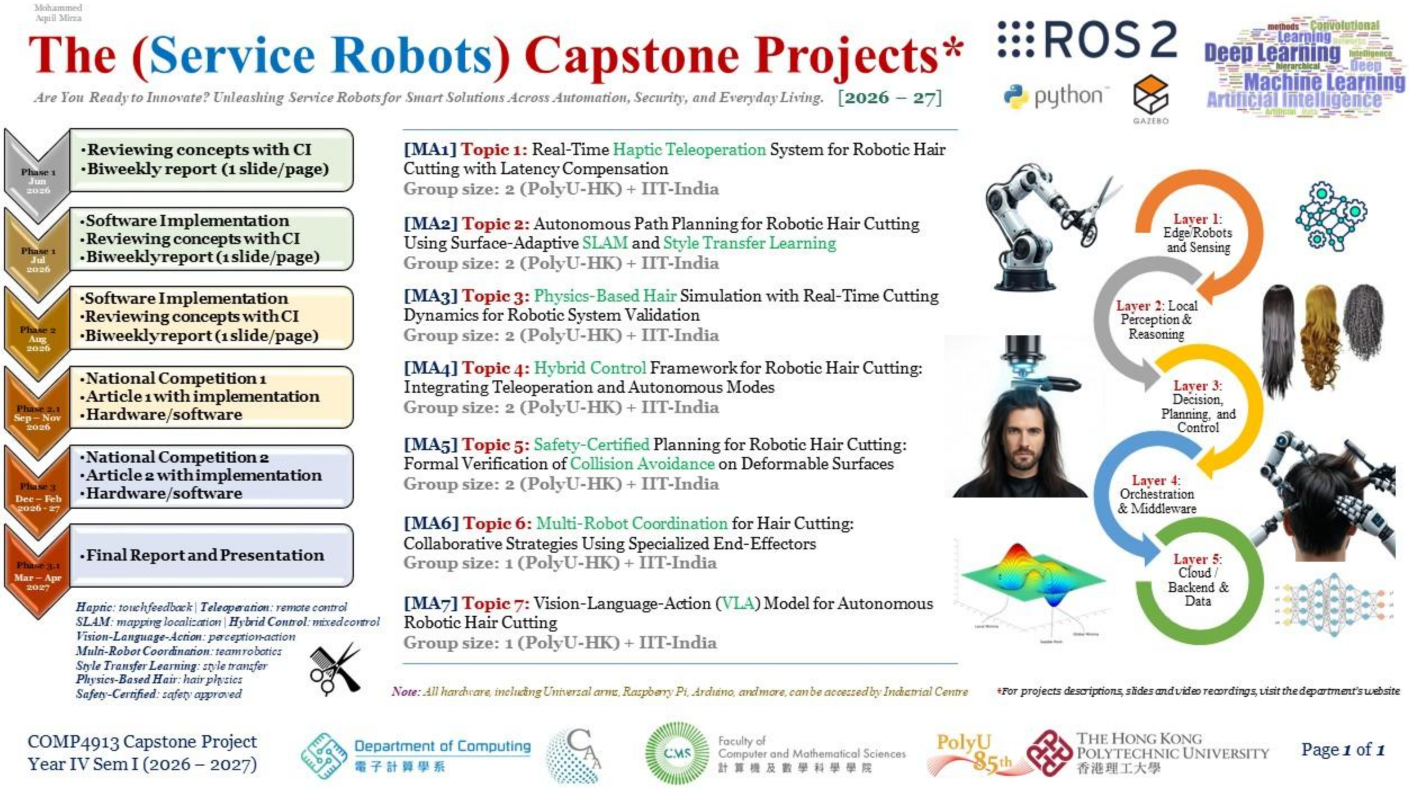
The Hong Kong Polytechnic University (PolyU), Hong Kong



MA1_Haptic | MA1 – Haptic Teleoperation System

2026-2027											
MA1: Real Time Haptic Teleoperation System for Robotic Hair Cutting with Latency Compensation											
The Hong Kong Polytechnic University (PolyU), Hong Kong											
Dr. Mohammed Aquil Mirza											
Task	**Description**	**M1**	**M2**	**M3**	**M4**	**M5**	**M6**	**M7**	**M8**	**M9**	**M10**
Foundations	ROS, Python, Haptics basics										
Device Integration	Haptic device setup										
Force Feedback	Basic force rendering										
Teleoperation	Master-slave control										
Latency Study	Analyze delay										
Predictive Control	Latency compensation										
3D UI	Visualization system										
System Integration	Combine modules										
Testing	User tests										
Optimization	Improve stability										

Month 1: Foundations – *ROS, Python, Haptics Basics*
Week 1 - Install ROS and configure the development environment; verify all packages and dependencies
Week 2 - Complete Python programming fundamentals with a focus on robotics scripting and ROS nodes
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Study haptic systems theory: force rendering, impedance control, and human–robot interaction principles
Week 4 - Run basic ROS nodes end-to-end and validate haptic system communication and sensor readings
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 2: Device Integration – *Haptic Device Setup*
Week 1 - Review haptic device hardware specifications, connectors, and interface options
Week 2 - Install device drivers, SDK libraries, and all software dependencies
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Establish device–computer communication and run diagnostic tests on device responses
Week 4 - Integrate device with ROS using custom node wrappers and perform initial calibration
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 3: Force Feedback – *Basic Force Rendering*
Week 1 - Study force rendering algorithms: spring–mass models, virtual walls, and impedance rendering
Week 2 - Implement basic constant-force and spring-force rendering modes in code
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Test force rendering against simple virtual objects in simulation and measure accuracy
Week 4 - Tune force parameters for stability and evaluate rendering quality across test scenarios
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 4: Teleoperation – *Master-Slave Control*
Week 1 - Study teleoperation architectures: position–position, force–position, and bilateral control
Week 2 - Configure master and slave robot hardware and simulation counterparts
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Implement position mirroring and velocity tracking between master and slave
Week 4 - Close the teleoperation loop, run tracking tests, and measure position error metrics
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 5: Latency Study – *Analyze Delay*
Week 1 - Design latency measurement methodology: define measurement points and benchmarking protocol
Week 2 - Measure round-trip delay across all system components (sensing, computation, actuation)
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Identify and rank the primary latency sources in the teleoperation pipeline
Week 4 - Document the full latency profile and propose targeted compensation strategies
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 6: Predictive Control – *Latency Compensation*
Week 1 - Review predictive control theory: Smith predictors, model-based prediction, and Kalman filtering
Week 2 - Implement a prediction model for estimating future robot state given measured delay
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Integrate the predictor into the teleoperation control loop
Week 4 - Compare closed-loop performance with and without latency compensation across delay conditions
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 7: 3D UI – *Visualization System*
Week 1 - Define 3D visualization requirements: data streams, frame rate targets, and UI layout design
Week 2 - Build a real-time 3D rendering environment displaying robot state and workspace
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Add haptic overlay visualizations: force vectors, contact regions, and safety zones
Week 4 - Conduct usability testing of the 3D interface and apply refinements based on feedback
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 8: System Integration – *Combine Modules*
Week 1 - Map all module interfaces and define data contracts between subsystems
Week 2 - Integrate haptic device, teleoperation controller, and force feedback layers
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Connect predictive control and 3D UI into the unified system pipeline
Week 4 - Perform end-to-end integration testing, log errors, and resolve compatibility issues
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 9: Testing – *User Tests*
Week 1 - Design user study protocol: tasks, metrics (task time, error, workload), and consent procedures
Week 2 - Recruit participants and conduct pilot sessions to validate the study design
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Run full user testing sessions with the integrated haptic teleoperation system
Week 4 - Analyse quantitative and qualitative results and document key findings and improvements
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 10: Optimization – *Improve Stability*
Week 1 - Profile system performance under load and identify computational and control bottlenecks
Week 2 - Optimize latency-critical components: reduce overhead in ROS callbacks and control loops
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Implement stability enhancements: damping tuning, noise filtering, and control margin adjustments
Week 4 - Run final validation tests and compile the optimized system documentation and benchmarks
Submit a biweekly report on your achievements and your plan for the next two weeks

***** The End *****

2026 – 2027 The Capstone Project – Robotic Hair Cutting System – Monthly Planning Breakdown
The Hong Kong Polytechnic University (PolyU), Hong Kong

MA2_SLAM_Path | MA2 – SLAM-Based Path Planning for Robotic Hair Cutting

2026-2027											
MA2: Autonomous Path Planning for Robotic Hair Cutting Using Surface Adaptive SLAM and Style Transfer Learning											
The Hong Kong Polytechnic University (PolyU), Hong Kong											
Dr. Mohammed Aquil Mirza											
Task	**Description**	**M1**	**M2**	**M3**	**M4**	**M5**	**M6**	**M7**	**M8**	**M9**	**M10**
Foundations	Install ROS, simulator, Python basics, run simple robot motion	■									
3D Basics	Load and visualize 3D head models and point clouds		■								
Head Model Setup	Convert mesh to point cloud and align coordinate systems		■	■							
Surface Analysis	Compute normals and curvature of head surface			■	■						
Region Segmentation	Define safe zones like ears, eyes, and neck				■						
Basic Path Planning	Generate simple coverage paths on surface					■					
Curved Surface Paths	Adapt paths to curved head geometry						■				
Style Mapping	Map hairstyle input to cutting regions/lengths						■	■			
Path Optimization	Reduce redundant paths and smooth trajectories							■			
Robot Execution	Execute planned paths in simulation								■		
Tool Orientation	Align cutting tool with surface normals								■		
Safety System	Add collision avoidance and emergency stops									■	
Integration	Combine perception, planning, and execution modules									■	■
Evaluation & Demo	Test system and prepare final demo										■

Month 1: Foundations – *Install ROS, Simulator, Python Basics, Run Simple Robot*
Week 1 - Install ROS and configure the Gazebo/simulation environment with all required packages
Week 2 - Complete Python basics for robotics: scripting, ROS topic pub/sub, and service calls
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Launch and manually control a simple robot arm in simulation using ROS
Week 4 - Read and log basic sensor data (camera, LiDAR) from the simulated robot in ROS
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 2: 3D Basics – *Load and Visualize 3D Head Models and Point Clouds*
Week 1 - Study common 3D data formats: PLY, OBJ, STL; understand mesh and point cloud structures
Week 2 - Load and display 3D head models using Open3D and/or PCL libraries
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Visualize point clouds in RViz and analyse density, noise, and coordinate systems
Week 4 - Apply basic 3D transformations and verify coordinate frame consistency across tools
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 3: Head Model Setup – *Convert Mesh to Point Cloud and Align Coordinate System*
Week 1 - Study mesh-to-point-cloud conversion techniques: uniform sampling, Poisson-disk sampling
Week 2 - Implement mesh sampling pipeline to generate clean, dense point clouds from head mesh
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Define a canonical coordinate system for the head model and implement alignment
Week 4 - Validate alignment accuracy quantitatively and handle edge cases such as incomplete meshes
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 4: Surface Analysis – *Compute Normals and Curvature of Head Surface*
Week 1 - Study surface normal estimation methods: PCA-based normals, orientation consistency
Week 2 - Implement normal computation on the head point cloud and visualize orientation fields
Submit a biweekly report on your achievements and your plan for the next two weeks

Week 3 - Compute principal curvature using eigenvalue analysis at each surface point
Week 4 - Validate normal and curvature maps against known geometric regions of the head model
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 5: Region Segmentation – *Define Safe Zones (Ears, Eyes, Neck)*
Week 1 - Study anatomical landmarks relevant to robotic hair cutting safety constraints
Week 2 - Implement landmark detection or manual annotation pipeline on the head model
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Define and label restricted zones (ears, eyes, neck) and cutting regions
Week 4 - Validate segmentation accuracy with visual inspection and quantitative boundary metrics
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 6: Basic Path Planning – *Generate Simple Coverage Paths on Surface*
Week 1 - Study coverage path planning algorithms: boustrophedon, spiral, and lawnmower patterns
Week 2 - Implement basic zigzag coverage path generation for flat surfaces
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Project coverage paths onto the 3D head mesh surface
Week 4 - Evaluate coverage completeness, path continuity, and gap percentage across test models
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 7: Curved Surface Paths – *Adapt Paths to Curved Head Geometry*
Week 1 - Study geodesic path planning and surface-aware trajectory methods
Week 2 - Implement geodesic path projection respecting head surface curvature
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Adapt path step sizes locally based on curvature magnitude to maintain uniform coverage
Week 4 - Test and visualize adapted paths on the full head model and measure coverage fidelity
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 8: Style Mapping – *Map Hairstyle Input to Cutting Regions/Lengths*
Week 1 - Design a structured hairstyle input schema specifying cutting zones and target lengths
Week 2 - Map textual or categorical hairstyle descriptors to segmented head regions
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Assign cutting length and direction parameters to each segmented region
Week 4 - Validate style-to-region mapping with a set of representative hairstyle specifications
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 9: Path Optimization – *Reduce Redundant Paths and Smooth Trajectories*
Week 1 - Identify redundant path segments, overlapping strokes, and unnecessary direction changes
Week 2 - Implement path pruning and segment merging to minimise redundancy
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Apply spline-based smoothing to waypoint sequences for smoother robot motion
Week 4 - Benchmark optimized paths against the original baseline on coverage, length, and smoothness
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 10: Robot Execution – *Execute Planned Paths in Simulation*
Week 1 - Configure the robot arm simulation (Gazebo + MoveIt) for path execution
Week 2 - Convert planned surface paths to robot joint trajectories using MoveIt
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Execute trajectories in simulation and monitor Cartesian and joint-space behaviour
Week 4 - Debug trajectory failures, adjust speed profiles, and validate path following accuracy
Submit a biweekly report on your achievements and your plan for the next two weeks

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Month 11: Tool Orientation – *Align Cutting Tool with Surface Normals*
Week 1 - Study tool orientation constraints for cutting tools in contact with curved surfaces
Week 2 - Compute desired tool frame orientations from surface normal vectors at each waypoint
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Integrate tool orientation planning into the MoveIt trajectory generation pipeline
Week 4 - Validate tool alignment consistency across the full head surface path
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 12: Safety System – *Add Collision Avoidance and Emergency Stops*
Week 1 - Define safety zones, force thresholds, and collision detection parameters
Week 2 - Implement real-time collision detection using simulation physics and contact sensors
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Add emergency stop triggers and safe retraction behaviours on violation detection
Week 4 - Stress-test the safety system with adversarial and edge-case simulation scenarios
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 13: Integration – *Combine Perception, Planning, and Execution Modules*
Week 1 - Define integration interfaces and data contracts between all pipeline modules
Week 2 - Connect perception output (head model, segmentation) to the path planning input
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Link path planning output to robot execution with live safety monitoring
Week 4 - Run end-to-end integration tests across the full perception–planning–execution pipeline
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 14: Evaluation & Demo – *Test System and Prepare Final Demo*
Week 1 - Design evaluation protocol: define success criteria for coverage, safety, and accuracy
Week 2 - Run quantitative tests measuring path coverage, cutting accuracy, and safety events
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Prepare the final demonstration setup: scenario scripts, visualizations, and documentation
Week 4 - Deliver the final demonstration and compile the comprehensive evaluation report
Submit a biweekly report on your achievements and your plan for the next two weeks

***** The End *****

2026 – 2027 The Capstone Project – Robotic Hair Cutting System – Monthly Planning Breakdown
The Hong Kong Polytechnic University (PolyU), Hong Kong

MA3_PhysiscSim | MA3 – Physics-Based Hair Simulation with Real-Time Cutting Dynamics

2026-2027											
MA3: Physics Based Hair Simulation with Real Time Cutting Dynamics for Robotic System Validation											
The Hong Kong Polytechnic University (PolyU), Hong Kong											
Dr. Mohammed Aquil Mirza											
Task	**Description**	**M1**	**M2**	**M3**	**M4**	**M5**	**M6**	**M7**	**M8**	**M9**	**M10**
Foundations	Physics + simulation basics										
Population Models	Hair diversity modeling										
Strand Dynamics	Forces & interactions										
Engine Integration	PhysX/Bullet setup										
Dynamic Interaction	Strand disconnection										
Mechanical Impact	Clipper interaction										
Real-Time System	Optimize simulation										
Data Collection	Capture behaviors										
Validation	Benchmarking										
Final Output	Demo + results										

Month 1: Foundations – *Physics + Simulation Basics*

Week 1
Install the simulation stack and configure Python/C++ build tools for the selected engine.
Review rigid body, particle, and spring-mass simulation basics relevant to hair modeling
Week 2
Study the structure of an existing physics engine pipeline and identify where hair modules plug in.
Run a small demo scene and log frame rate, timestep, and collision debug information.
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3
Create a minimal project scene with a scalp mesh, a simple strand proxy, and a movable tool object.
Document the software architecture, dependencies, and expected data flow for the full simulator.
Week 4
Validate that the development environment can load assets, run physics steps, and export diagnostic outputs.
Prepare a short technical note describing initial setup issues and how they were resolved.
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 2: Population Models – *Hair Diversity Modeling*

Week 1
Study how hair populations differ in density, strand thickness, curl pattern, and growth direction.
Define parameter sets for at least three representative hair profiles: straight, wavy, and curly.
Week 2
Implement a parameter schema storing strand count, length distribution, curl radius, and stiffness.
Generate synthetic scalp regions with different density maps and strand seeding patterns.
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3
Visualize parameterized populations and compare appearance changes under different distributions.
Refine the generation rules so that strand placement remains realistic near the hairline and crown.

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Week 4
Benchmark generation time and memory cost for the selected hair population settings.
Freeze a reusable library of validated population presets for later simulation experiments.
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 3: Strand Dynamics – *Forces and Interactions*
Week 1
Review the physical forces acting on hair strands: gravity, elasticity, damping, and friction.
Implement per-strand state updates for position, velocity, and constraint enforcement.
Week 2
Add inter-strand interaction logic such as simplified collision handling or proximity response.
Evaluate stability under different time steps and integrators such as Euler or semi-implicit Euler.
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3
Test strand motion for straight and curly profiles under gravity and external motion disturbance.
Measure oscillation decay, numerical stability, and qualitative realism across force settings.
Week 4
Tune stiffness and damping coefficients to avoid jitter while preserving believable motion.
Document a validated dynamic model ready for engine-level integration.
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 4: Engine Integration – *PhysX/Bullet Setup*
Week 1
Compare Bullet, PhysX, or the chosen engine APIs for soft-body or particle-based hair simulation support.
Select the integration approach and define data exchange between hair solver and rendering layer.
Week 2
Wrap the strand simulation module inside the engine update loop.
Implement import/export of scalp meshes, strand parameters, and collision proxies.
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3
Connect debug visualization for strand segments, collision points, and timing statistics.
Run integration tests to verify the solver advances correctly inside the engine timestep.
Week 4
Profile runtime cost per simulation step and identify bottlenecks in collision or constraint handling.
Stabilize the integrated module for real-time tests with moderate strand counts.
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 5: Dynamic Interaction – *Strand Disconnection*
Week 1
Study how cutting events should be modeled: strand splitting, strand deletion, or topology updates.
Define event rules for when a cutting tool intersects a strand or group of strands.
Week 2
Implement collision checks between the cutting tool and strand geometry.
Trigger strand disconnection logic while preserving solver stability after the cut event.
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3
Visualize cut strands before and after interaction to verify the event timing and geometry updates.
Test repeated cuts under different tool speeds and approach angles.
Week 4
Measure event latency and post-cut simulation stability under multiple interaction conditions.
Refine the cutting event system so it can support repeated real-time grooming actions.

Submit a biweekly report on your achievements and your plan for the next two weeks

Month 6: Mechanical Impact – *Clipper Interaction*

Week 1

Study the geometry and motion characteristics of different clipping mechanisms relevant to the simulator.

Create simplified tool models capturing approach angle, blade width, and contact region.

Week 2

Implement contact response between tool and hair bundle for push, sweep, and cutting interactions.

Model force transmission or deformation effect on nearby strands during tool passage.

Submit a biweekly report on your achievements and your plan for the next two weeks

Week 3

Run controlled experiments varying clipper speed, pressure proxy, and direction.

Compare the resulting strand behavior and cut patterns across multiple settings.

Week 4

Refine tool parameters to produce stable and interpretable interaction outcomes.

Document the final interaction model and assumptions used in the simulation.

Submit a biweekly report on your achievements and your plan for the next two weeks

Month 7: Real-Time System – *Optimize Simulation*

Week 1

Profile the full pipeline and identify the slowest modules: collision checks, solver steps, or rendering.

Define real-time targets for frame rate and per-frame physics budget.

Week 2

Introduce level-of-detail or adaptive update strategies for inactive or distant strand groups.

Optimize memory usage and data layout for faster solver updates.

Submit a biweekly report on your achievements and your plan for the next two weeks

Week 3

Parallelize suitable components or reduce unnecessary debug overhead during runtime.

Compare performance before and after optimization using repeatable benchmark scenes.

Week 4

Run stress tests at increasing strand counts and record the point at which real-time performance drops.

Freeze an optimized configuration for downstream validation experiments.

Submit a biweekly report on your achievements and your plan for the next two weeks

Month 8: Data Collection – *Capture Behaviors*

Week 1

Define what data must be logged: strand motion statistics, cut events, timing, tool trajectory, and contact states.

Create a logging format for reproducible benchmarking runs.

Week 2

Run simulations across multiple hair types and interaction settings to build a validation dataset.

Capture synchronized visual outputs and numeric logs for later analysis.

Submit a biweekly report on your achievements and your plan for the next two weeks

Week 3

Organize the dataset by scenario, parameter setting, and observed behavior class.

Check data completeness and remove corrupted or inconsistent runs.

Submit a biweekly report on your achievements and your plan for the next two weeks

Week 4

Produce descriptive statistics and representative visual examples for each condition.

Prepare the cleaned dataset package for validation and reporting.

Submit a biweekly report on your achievements and your plan for the next two weeks

Month 9: Validation – *Benchmarking*

Week 1

Define validation metrics such as cut accuracy proxy, frame rate, event consistency, and realism indicators.

Establish baseline scenes for comparison across model revisions.

Week 2

Run controlled benchmark experiments across hair profiles, tool settings, and solver variants.

Compare simulation outcomes against expected mechanical behavior and internal consistency checks.

Submit a biweekly report on your achievements and your plan for the next two weeks

Week 3

Analyse numerical results and inspect representative videos or rendered outputs for qualitative validation.

Identify the strongest and weakest conditions in the current simulator.

Week 4

Summarize benchmark findings and translate them into concrete refinement priorities.

Prepare evaluation plots, tables, and concise interpretation notes.

Submit a biweekly report on your achievements and your plan for the next two weeks

Month 10: Final Output – *Demo + Results*

Week 1

Integrate all validated components into a polished demonstration scene.

Prepare a short scenario script showing diversity modeling, dynamic cutting, and benchmark outputs.

Week 2

Run end-to-end tests to ensure stable performance throughout the full demonstration pipeline.

Create final figures, screenshots, and a compact explanation of the system architecture.

Submit a biweekly report on your achievements and your plan for the next two weeks

Week 3

Compile quantitative results, qualitative findings, limitations, and lessons learned.

Refine documentation so a new student can reproduce the complete simulator setup.

Week 4

Deliver the final demo and archive all source code, assets, and benchmark logs.

Submit the complete project report and checklist of reusable outputs for future work.

Submit a biweekly report on your achievements and your plan for the next two weeks

***** The End *****

MA4_HybridControl | MA4 – Hybrid Control Framework for Robotic Hair Cutting

2026-2027											
MA4: Hybrid Control Framework for Robotic Hair Cutting: Integrating Teleoperation and Autonomous Modes											
The Hong Kong Polytechnic University (PolyU), Hong Kong											
Dr. Mohammed Aquil Mirza											
Task	**Description**	**M1**	**M2**	**M3**	**M4**	**M5**	**M6**	**M7**	**M8**	**M9**	**M10**
Foundations	Control + ROS basics										
System Architecture	Design framework										
Task Segmentation	Divide haircut tasks										
Autonomous Mode	Auto execution										
Teleoperation Mode	Manual control										
Mode Switching	Switch logic										
Adaptive Interface	User feedback										
Hybrid Control	Combine modes										
Evaluation Metrics	Performance analysis										
Final Demo	Integrated system										

Month 1: Foundations – *Control + ROS Basics*

Week 1

Install ROS/ROS2, configure the robotics workspace, and verify controller package dependencies.

Review control fundamentals: feedback loops, stability, PID control, and state estimation.

Week 2

Run basic robot arm simulations and inspect sensor, actuator, and command interfaces.

Implement simple ROS nodes for reading robot state and sending low-level motion commands.

Submit a biweekly report on your achievements and your plan for the next two weeks

Week 3

Study how teleoperation and autonomous control are typically structured in shared-control systems.

Map the required subsystems for the hybrid control project: sensing, switching, UI, and safety.

Week 4

Validate that the environment supports real-time state updates and closed-loop control tests.

Document the initial controller stack and development roadmap for the hybrid framework.

Submit a biweekly report on your achievements and your plan for the next two weeks

Month 2: System Architecture – *Design Framework*

Week 1

Define the top-level architecture: teleoperation layer, autonomous planner, supervisor, and switching logic.

Specify module interfaces and data flow between sensing, decision, and actuation components.

Week 2

Draft block diagrams showing how commands are routed under each operating mode.

Select the middleware and message structure for status, intent, and control signals.

Submit a biweekly report on your achievements and your plan for the next two weeks

Week 3

Implement the base software skeleton with placeholder modules for each subsystem.

Create test hooks so each module can be validated independently before full integration.

Week 4

Review the architecture for timing risks, failure points, and extensibility.

Freeze version 1 of the framework specification and implementation skeleton.

Submit a biweekly report on your achievements and your plan for the next two weeks

Month 3: Task Segmentation – *Divide Haircut Tasks*

Week 1

Study how haircut procedures can be decomposed into atomic tasks, regions, and motion primitives.

Define a task taxonomy covering trimming, edge cleanup, contour following, and stylist-directed actions.

Week 2

Implement a representation for task segments including priority, difficulty, and preferred mode.

Map example hairstyle procedures into segmented task sequences.

Submit a biweekly report on your achievements and your plan for the next two weeks

Week 3

Associate each segment with constraints such as precision requirement, speed tolerance, and safety margin.

Test the segmentation logic on several representative haircut scenarios.

Week 4

Refine the segmentation scheme so it is consistent and machine-readable for controller decisions.

Prepare a task library that will drive mode switching in later months.

Submit a biweekly report on your achievements and your plan for the next two weeks

Month 4: Autonomous Mode – *Auto Execution*

Week 1

Study autonomous control strategies for repetitive, high-precision tasks on constrained surfaces.

Select the planner/controller combination to use in automatic mode.

Week 2

Implement baseline autonomous execution for a small subset of segmented haircut tasks.

Connect perception or predefined task inputs to the autonomous controller.

Submit a biweekly report on your achievements and your plan for the next two weeks

Week 3

Run trials to measure path tracking quality and robustness in straightforward scenarios.

Identify the task types where autonomous mode is strong enough to operate without human correction.

Week 4

Document the controller assumptions, limitations, and expected operating envelope.

Freeze the autonomous mode baseline for later hybrid integration.

Submit a biweekly report on your achievements and your plan for the next two weeks

Month 5: Teleoperation Mode – *Manual Control*

Week 1

Study operator-in-the-loop control principles and suitable interfaces for fine grooming manipulation.

Set up the teleoperation interface and map human input to robot action commands.

Week 2

Implement scaling, filtering, or motion constraints to improve manual precision.

Run test scenarios for delicate segments that benefit from operator judgement.

Submit a biweekly report on your achievements and your plan for the next two weeks

Week 3

Measure teleoperation responsiveness, tracking smoothness, and user effort.

Identify the situations where manual control outperforms autonomous execution.

Week 4

Refine the teleoperation mapping so it remains intuitive and safe near sensitive areas.

Prepare the teleoperation module for seamless use inside the hybrid framework.

Submit a biweekly report on your achievements and your plan for the next two weeks

Month 6: Mode Switching – *Switch Logic*

Week 1
Study supervisory control and state-machine design for switching between autonomous and manual modes.
Define trigger conditions for switching: uncertainty, task type, user override, and safety events.
Week 2
Implement the switching logic as a supervisor or state machine managing both control modes.
Ensure mode transitions preserve command continuity and do not create abrupt motion jumps.
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3
Test the switching mechanism under scripted scenarios with normal and edge-case transitions.
Measure switching delay, stability, and command consistency during handover.
Week 4
Refine transition conditions and fallback behaviors based on observed failure cases.
Lock the switching logic for integration with the adaptive interface and shared autonomy module.
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 7: Adaptive Interface – *User Feedback*

Week 1
Define what feedback the operator should receive during autonomous, manual, and switching phases.
Design interface elements for status, confidence, safety warnings, and suggested actions.
Week 2
Implement a user-facing dashboard or lightweight UI showing active mode and key system indicators.
Add visual or haptic notifications for takeover requests and safety constraints.
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3
Test the interface in interactive sessions and measure usability, clarity, and reaction time.
Identify UI elements that overload or distract the operator.
Week 4
Refine the interface so it supports fast understanding during mode transitions.
Finalize the adaptive interface specification for full system integration.
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 8: Hybrid Control – *Combine Modes*

Week 1
Integrate autonomous execution, teleoperation, and supervisory switching into one control pipeline.
Define shared state variables and arbitration rules for conflicting commands.
Week 2
Run combined-mode scenarios where the robot alternates between automatic and human-guided actions.
Measure continuity of motion and task completion quality across hybrid runs.
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3
Tune assistance level, authority blending, and timing to reduce unnecessary takeovers.
Stress-test the pipeline with noisy inputs, delayed feedback, and ambiguous tasks.
Week 4
Document the mature hybrid controller behavior and its performance envelope.
Prepare the integrated framework for final evaluation.
Submit a biweekly report on your achievements and your plan for the next two weeks

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Month 9: Evaluation Metrics – *Performance Analysis*

Week 1

Define quantitative metrics for autonomy quality, teleoperation quality, switching smoothness, and safety.

Create a repeatable benchmark protocol with representative task scenarios.

Week 2

Run comparative experiments for autonomous-only, teleoperation-only, and hybrid control modes.

Record success rate, task time, error, operator workload proxy, and number of interventions.

Submit a biweekly report on your achievements and your plan for the next two weeks

Week 3

Analyse results statistically and identify where hybrid control provides measurable benefit.

Prepare plots and summary tables comparing the three operating modes.

Week 4

Translate evaluation outcomes into clear conclusions, limitations, and recommended next steps.

Assemble the final metrics package for the report and demo.

Submit a biweekly report on your achievements and your plan for the next two weeks

Month 10: Final Demo – *Integrated System*

Week 1

Prepare a polished demonstration script showing task segmentation, switching, and shared autonomy behavior.

Run rehearsal sessions and fix remaining issues in timing, stability, and interface presentation.

Week 2

Produce final screenshots, diagrams, and a concise explanation of the hybrid architecture.

Verify that all modules launch reliably from a clean project setup.

Submit a biweekly report on your achievements and your plan for the next two weeks

Week 3

Deliver the final integrated demonstration and collect final feedback from supervisors or peers.

Archive code, configs, benchmark results, and operator instructions.

Week 4

Compile the final report with architecture, experiments, results, and future improvement points.

Submit the completed project package in a reusable student-friendly format.

Submit a biweekly report on your achievements and your plan for the next two weeks

***** The End *****

MA5_Safety | MA5 – Safety-Verified Motion Planning for Robotic Hair Cutting

2026-2027											
MA5: Safety Certified Planning for Robotic Hair Cutting: Formal Verification of Collision Avoidance on Deformable Surfaces											
The Hong Kong Polytechnic University (PolyU), Hong Kong											
Dr. Mohammed Aquil Mirza											
Task	**Description**	**M1**	**M2**	**M3**	**M4**	**M5**	**M6**	**M7**	**M8**	**M9**	**M10**
Foundations	Math + ROS basics										
3D Basics	Head modeling										
Surface Modeling	Curved geometry										
Deformable Modeling	Elastic surfaces										
Formal Methods	UPPAAL/PRISM										
Path Planning	Safe trajectories										
Replanning	Dynamic updates										
Collision Avoidance	Safety guarantees										
Verification	Formal validation										
Evaluation	Final system test										

Month 1: Foundations – *Math + ROS Basics*
Week 1 - Review relevant mathematics: linear algebra, probability theory, and differential geometry
Week 2 - Install ROS and configure the simulation environment for safety-critical robotics
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Study safety-critical systems theory, standards (ISO 10218), and failure mode analysis
Week 4 - Run baseline ROS nodes and validate the complete development environment setup
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 2: 3D Basics – *Head Modeling*
Week 1 - Study 3D geometric representations: meshes, voxels, signed-distance fields
Week 2 - Load and inspect 3D head models in simulation and RViz
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Review head anatomy critical to safe robot interaction (vulnerable zones, tissue properties)
Week 4 - Validate head model fidelity and confirm coordinate system consistency
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 3: Surface Modeling – *Curved Geometry*
Week 1 - Study parametric surface representations: NURBS, B-splines, and implicit surfaces
Week 2 - Implement mesh generation for curved scalp geometry from point cloud inputs
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Evaluate surface model accuracy using point-to-surface distance metrics
Week 4 - Document the surface modeling pipeline with inputs, outputs, and quality benchmarks
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 4: Deformable Modeling – *Elastic Surfaces*
Week 1 - Study elastic and deformable body simulation methods: spring-mass networks and FEM
Week 2 - Implement a spring-mass or finite element deformation model for scalp tissue
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Simulate scalp deformation under controlled applied forces and measure response
Week 4 - Validate deformation model against expected physical behaviour and material properties
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 5: Formal Methods – *UPPAAL/PRISM*
Week 1 - Introduction to formal verification: temporal logic, model checking, and safety specifications
Week 2 - Model the robot–environment interaction system in UPPAAL timed automata
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Specify safety properties (collision avoidance, force limits) and verify using PRISM
Week 4 - Interpret verification results, identify counterexamples, and refine system models
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 6: Path Planning – *Safe Trajectories*
Week 1 - Study safety-constrained motion planning: CBF-RRT, potential fields, and safe corridors
Week 2 - Implement a safety-constrained planner for generating head-surface trajectories
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Test the planner on the head model with defined force and proximity safety constraints
Week 4 - Evaluate trajectory safety compliance metrics and constraint satisfaction rates
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 7: Replanning – *Dynamic Updates*
Week 1 - Study online replanning methods: reactive planning, contingency trees, and MPC
Week 2 - Implement trigger conditions for dynamic replanning on disturbance detection
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Test the replanning module under simulated head movements and unexpected contact
Week 4 - Measure replanning latency and verify that safety guarantees are preserved post-replan
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 8: Collision Avoidance – *Safety Guarantees*
Week 1 - Study formal collision avoidance guarantees: control barrier functions and safety certificates
Week 2 - Implement control barrier function (CBF) filters on the robot control input
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Integrate CBF-based collision avoidance into the full planning and execution stack
Week 4 - Verify safety guarantees through simulation stress tests and statistical analysis
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 9: Verification – *Formal Validation*
Week 1 - Define formal verification test suite covering all specified safety properties
Week 2 - Run model checking on the integrated system using UPPAAL and PRISM
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Conduct simulation-based validation to complement formal proofs with empirical evidence
Week 4 - Document verification outcomes, resolve any property violations, and update models
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 10: Evaluation – *Final System Test*
Week 1 - Design a comprehensive evaluation protocol with quantitative success criteria
Week 2 - Run end-to-end system tests across diverse head geometries and motion scenarios
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Analyse results against safety benchmarks and compare with baseline planners
Week 4 - Compile the final evaluation report and deliver the complete system demonstration
Submit a biweekly report on your achievements and your plan for the next two weeks

***** The End *****

MA6_MultiRobot | MA6 – Multi-Robot Coordination for Autonomous Hair Cutting

2026-2027											
MA6: Multi Robot Coordination for Hair Cutting: Collaborative Strategies Using Specialized End Effectors											
The Hong Kong Polytechnic University (PolyU), Hong Kong											
Dr. Mohammed Aquil Mirza											
Task	**Description**	**M1**	**M2**	**M3**	**M4**	**M5**	**M6**	**M7**	**M8**	**M9**	**M10**
Foundations	Multi-robot basics										
Kinematics	Robot motion										
Task Allocation	Assign tasks										
Coordination	Multi-robot sync										
Collision Avoidance	Multi-agent safety										
Simulation	Environment setup										
Optimization	Efficiency tuning										
Human-Robot Interaction	User integration										
Testing	System validation										
Final Demo	Complete pipeline										

Month 1: Foundations – *Multi-Robot Basics*
Week 1 - Study multi-robot systems: architectures, communication patterns, and coordination paradigms
Week 2 - Install ROS multi-robot simulation environment and configure namespacing for multiple agents
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Launch and individually control multiple simulated robots in the same environment
Week 4 - Test inter-robot communication: topic publishing, service calls, and synchronization primitives
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 2: Kinematics – *Robot Motion*
Week 1 - Review forward and inverse kinematics for robotic arms used in the project
Week 2 - Implement kinematic solvers for each robot in the multi-robot team
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Validate kinematic solutions by comparing computed and simulated end-effector poses
Week 4 - Analyse individual robot workspace coverage and identify joint limit constraints
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 3: Task Allocation – *Assign Tasks*
Week 1 - Study task allocation algorithms: auction-based, market-inspired, and combinatorial methods
Week 2 - Define task representation schema including location, duration, and precedence constraints
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Implement a baseline task allocation algorithm and integrate with the multi-robot setup
Week 4 - Test allocation performance under varying task loads and measure makespan and fairness
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 4: Coordination – *Multi-Robot Sync*
Week 1 - Study synchronization protocols: barrier synchronization, token passing, and consensus methods
Week 2 - Implement a communication-based coordination layer between robot agents
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Test synchronized task execution by running coordinated motions across all robots
Week 4 - Measure coordination overhead, timing jitter, and synchronization accuracy
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 5: Collision Avoidance – *Multi-Agent Safety*
Week 1 - Study multi-agent collision avoidance: ORCA, velocity obstacles, and CBS planning
Week 2 - Implement a multi-agent collision avoidance module for the robot team
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Test avoidance in high-density simulation scenarios with overlapping robot workspaces
Week 4 - Measure safety rate, false avoidance frequency, and impact on task completion time
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 6: Simulation – *Environment Setup*
Week 1 - Design a realistic simulation environment representing the hair cutting workspace
Week 2 - Add physical obstacles, workspace boundaries, and constraint zones to the simulation
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Integrate all robots, sensors, and end-effectors into the unified simulation
Week 4 - Validate environment physics accuracy and verify all sensor outputs are realistic
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 7: Optimization – *Efficiency Tuning*
Week 1 - Profile system performance: measure CPU load, inter-robot latency, and task throughput
Week 2 - Optimize task allocation for minimum makespan using improved scheduling heuristics
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Tune motion planning speed and safety margins for each robot to reduce idle time
Week 4 - Benchmark the optimized system against the baseline and document performance gains
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 8: Human-Robot Interaction – *User Integration*
Week 1 - Study HRI design principles for close-proximity collaborative robots
Week 2 - Implement a user interface for hairstyle task input, monitoring, and manual override
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Test human–robot interaction scenarios with simulated operator actions
Week 4 - Refine the interface based on usability evaluation and operator feedback
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 9: Testing – *System Validation*
Week 1 - Design a comprehensive validation test suite with unit and integration test cases
Week 2 - Run unit tests on each module: allocation, coordination, avoidance, and control
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Conduct full integration tests on the complete multi-robot pipeline end-to-end
Week 4 - Document test outcomes, resolve outstanding issues, and re-test until acceptance criteria are met
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 10: Final Demo – *Complete Pipeline*
Week 1 - Prepare demo scenario: define hairstyle tasks, robot roles, and demonstration sequence
Week 2 - Rehearse the full multi-robot pipeline and refine timing and visual presentation
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Run the final demonstration with the complete coordinated multi-robot system
Week 4 - Compile the final project report including architecture, results, and lessons learned
Submit a biweekly report on your achievements and your plan for the next two weeks

***** The End *****

MA7_VLA | MA7 – Vision-Language Action (VLA) Model for Robotic Hair Cutting

2026-2027
MA7: Vision Language Action (VLA) Model for Autonomous Robotic Hair Cutting
The Hong Kong Polytechnic University (PolyU), Hong Kong
Dr. Mohammed Aquil Mirza

Task	Description	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10
Foundations	Setup Python, PyTorch and basic ML understanding	■									
CLIP/VLM Intro	Use pretrained vision-language models for image-text tasks		■								
Segmentation	Detect hair regions using pretrained models			■							
Feature Extraction	Extract useful visual features from input images			■	■						
Action Space Definition	Define robot actions like move/cut operations				■						
Rule-Based Mapping	Create baseline mapping from input to actions					■					
Dataset Creation	Generate synthetic data for training					■	■				
Model Training	Train simple model to predict actions						■				
Multimodal Fusion	Combine image and text inputs							■			
Simulation Control	Convert model outputs into robot commands							■	■		
Closed Loop Control	Enable feedback-based execution								■		
Feedback Loop	Refine actions based on system state									■	
Safety Layer	Add confidence thresholds and safe stopping									■	■
Integration & Demo	Full system integration and demonstration										■

Month 1: Foundations – *Setup Python, PyTorch, and Basic ML Understanding*
Week 1 - Set up the Python development environment and install PyTorch with GPU support
Week 2 - Review foundational machine learning concepts: loss functions, gradient descent, and overfitting
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Implement simple ML models in PyTorch: logistic regression and basic neural networks
Week 4 - Train and evaluate a multi-layer perceptron on a benchmark dataset to validate the setup
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 2: CLIP/VLM Intro – *Pretrained Vision-Language Models for Image-Text Tasks*
Week 1 - Study the CLIP architecture: contrastive learning, image encoders, and text encoders
Week 2 - Load a pretrained CLIP model and run zero-shot image-text similarity tasks
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Explore CLIP embeddings using hairstyle images and natural language style descriptions
Week 4 - Evaluate CLIP performance on domain-relevant tasks and document findings
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 3: Segmentation – *Detect Hair Regions Using Pretrained Models*
Week 1 - Study semantic segmentation architectures: SAM, SegFormer, and Mask2Former
Week 2 - Run a pretrained segmentation model on hair and scalp images out-of-the-box
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Fine-tune or prompt-adapt the model to improve hair region detection accuracy
Week 4 - Evaluate segmentation performance with IoU and Dice metrics on a domain test set
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 4: Feature Extraction – *Extract Useful Visual Features from Input Images*
Week 1 - Study feature extraction from CNN backbones and Vision Transformers (ViT)
Week 2 - Extract visual features from hair images using a pretrained encoder backbone
Submit a biweekly report on your achievements and your plan for the next two weeks

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Week 3 - Analyse extracted features for sensitivity to hair texture, density, and structure
Week 4 - Benchmark feature quality by evaluating downstream task performance with extracted representations
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 5: Action Space Definition – *Define Robot Actions Like Move/Cut Operations*
Week 1 - Study robot action representations in imitation learning and policy gradient methods
Week 2 - Define the discrete and continuous action spaces covering all required cutting operations
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Implement action space encoding (one-hot, continuous vector) and decoding functions
Week 4 - Validate action space completeness by mapping all required robot motions to action representations
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 6: Rule-Based Mapping – *Create Baseline Mapping from Input to Actions*
Week 1 - Study rule-based control systems: decision trees, lookup tables, and finite state machines
Week 2 - Define deterministic mapping rules from hairstyle descriptors to robot action sequences
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Implement and test the rule-based controller across representative hairstyle inputs
Week 4 - Evaluate baseline controller performance to establish a benchmark for the learned model
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 7: Dataset Creation – *Generate Synthetic Data for Training*
Week 1 - Design the synthetic data generation pipeline: simulation parameters and output format
Week 2 - Generate paired image–action sequences for diverse hairstyle scenarios in simulation
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Apply data augmentation (lighting, pose variation, noise) to improve dataset diversity
Week 4 - Validate dataset quality and class balance; produce dataset statistics and example visualizations
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 8: Model Training – *Train Simple Model to Predict Actions*
Week 1 - Design a neural network architecture for action prediction from visual and language features
Week 2 - Set up the training pipeline: loss function, optimizer, scheduler, and logging
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Train the model on the synthetic dataset and monitor convergence metrics
Week 4 - Evaluate training performance with held-out validation data and tune hyperparameters
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 9: Multimodal Fusion – *Combine Image and Text Inputs*
Week 1 - Study multimodal fusion strategies: early fusion, late fusion, and cross-attention mechanisms
Week 2 - Implement a fusion module combining CLIP image and text embedding streams
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Train the multimodal model on the synthetic dataset with combined input features
Week 4 - Compare multimodal versus unimodal model performance on action prediction accuracy
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 10: Simulation Control – *Convert Model Outputs into Robot Commands*
Week 1 - Study model-to-robot command translation: output decoding and trajectory conversion
Week 2 - Implement decoding functions that map model output vectors to executable robot commands
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Execute predicted robot commands in simulation and measure Cartesian accuracy
Week 4 - Debug decoding failures, refine the translation pipeline, and re-evaluate accuracy
Submit a biweekly report on your achievements and your plan for the next two weeks

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Month 11: Closed Loop Control – *Enable Feedback-Based Execution*
Week 1 - Study closed-loop control architectures for VLA systems: observe–predict–act cycles
Week 2 - Implement the perception–action loop integrating the trained model into the control cycle
Submit a biweekly report on your achievements and your plan for the next two weeks

Week 3 - Test closed-loop execution across diverse hairstyle scenarios and environment conditions
Week 4 - Measure loop latency, action consistency, and control stability metrics
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 12: Feedback Loop – *Refine Actions Based on System State*
Week 1 - Design a state estimation pipeline to capture current progress and deviation from goal
Week 2 - Implement a state-conditioned action correction mechanism using system state feedback
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Test feedback-based refinement in simulation under disturbance and partial failure conditions
Week 4 - Quantify improvement in task completion rate versus the open-loop baseline
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 13: Safety Layer – *Add Confidence Thresholds and Safe Stopping*
Week 1 - Study safety mechanisms for learned robot controllers: uncertainty estimation and safe sets
Week 2 - Implement confidence score estimation for model output predictions at each timestep
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Add safe stopping and fallback behaviour triggered when confidence drops below threshold
Week 4 - Stress-test the safety layer with adversarial inputs and out-of-distribution scenarios
Submit a biweekly report on your achievements and your plan for the next two weeks

Month 14: Integration & Demo – *Full System Integration and Demonstration*
Week 1 - Integrate all modules: perception, VLM, action prediction, feedback, and safety layer
Week 2 - Run comprehensive end-to-end system tests and resolve all integration issues
Submit a biweekly report on your achievements and your plan for the next two weeks
Week 3 - Prepare the final demonstration: scripts, visualizations, and evaluation metrics summary
Week 4 - Deliver the final system demonstration and compile the complete project report
Submit a biweekly report on your achievements and your plan for the next two weeks

***** The End *****